(vi) $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha} = \frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}$

$= \frac{(-343)-3\left(10\times(-7)\right)}{10} = \frac{-343+210}{10} = \frac{-133}{10}$

Example 3.45 If α, β are the roots of the equation $3x^2+7x-2=0$, find the values of

(i) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$ (ii) $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}$

Solution $3x^2+7x-2=0$ here, $a=3$, $b=7$, $c=-2$

since, α, β are the roots of the equation

(i) $\alpha+\beta = \frac{-b}{a} = \frac{-7}{3}$; $\alpha\beta = \frac{c}{a} = \frac{-2}{3}$

$\frac{\alpha}{\beta}+\frac{\beta}{\alpha} = \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta} = \frac{\left(\frac{-7}{3}\right)^2-2\left(\frac{-2}{3}\right)}{\frac{-2}{3}} = \frac{-61}{6}$

(ii) $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha} = \frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta} = \frac{\left(-\frac{7}{3}\right)^3-3\left(-\frac{2}{3}\right)\left(-\frac{7}{3}\right)}{-\frac{2}{3}} = \frac{469}{18}$

Example 3.46 If α, β are the roots of the equation $2x^2-x-1=0$, then form the equation whose roots are (i) $\frac{1}{\alpha}, \frac{1}{\beta}$ (ii) $\alpha^2\beta$, $\beta^2\alpha$ (iii) $2\alpha+\beta$, $2\beta+\alpha$

Solution $2x^2-x-1=0$ here, $a=2$, $b=-1$, $c=-1$

$\alpha+\beta = \frac{-b}{a} = \frac{-(-1)}{2} = \frac{1}{2}$; $\alpha\beta = \frac{c}{a} = -\frac{1}{2}$

(i) Given roots are $\frac{1}{\alpha}, \frac{1}{\beta}$

Sum of the roots $= \frac{1}{\alpha}+\frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{\frac{1}{2}}{-\frac{1}{2}} = -1$

Product of the roots $= \frac{1}{\alpha}\times\frac{1}{\beta} = \frac{1}{\alpha\beta} = \frac{1}{-\frac{1}{2}} = -2$

The required equation is x^2 –(Sum of the roots)x + (Product of the roots)=0

$x^2-(-1)x-2=0 \Rightarrow x^2+x-2=0$

(ii) Given roots are $\alpha^2\beta$, $\beta^2\alpha$

Sum of the roots $\alpha^2\beta+\beta^2\alpha = \alpha\beta(\alpha+\beta) = -\frac{1}{2}\left(\frac{1}{2}\right) = -\frac{1}{4}$

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